

STANDARD OPERATING PROCEDURE

Taypro Fingerprint Attendance Device

Raspberry Pi 3B + R307S + OLED + LEDs

This document explains end-to-end how to configure the device, enroll employees, run daily attendance, and maintain the system at remote sites.

Audience: Site technicians, HR admins, IT support

System: device-to-erp (Pi) + taypro-console (HR)

1. Purpose & overview

The fingerprint attendance device replaces RFID card tapping with fingerprint scans. When an enrolled finger is placed on the R307S sensor, the device sends a punch to the Taypro cloud over MQTT. The OLED shows status for remote sites (no monitor required). HR manages employees and enrollment from the console.

1.1 What the device does

- Connects to office Wi-Fi / Ethernet and MQTT cloud
- Registers itself to HR as an attendance device
- Matches finger templates stored on the R307S sensor
- Publishes check-in / check-out punches to the server
- Shows results on OLED + 3 status LEDs
- Supports remote enroll from HR Admin UI (2 fingers per user)

1.2 Important storage rule

Fingerprint biometric templates live ON the R307S sensor chip (up to 1000). MongoDB / HR only stores mapping IDs like FP0001 and FP0002 on the employee record. If you replace the sensor hardware, users must be re-enrolled.

2. Hardware bill of materials

Item	Role
Raspberry Pi 3 Model B	Runs attendance software
R307 / R307S fingerprint sensor	Scan & store templates
CP2102 USB-UART converter	Sensor serial link
1.3" I2C OLED (SH1106/SSD1306)	On-site display
3x LEDs + 220 ohm resistors	Net / OK / Fail status
5V power for Pi + sensor VCC	Stable power (sensor needs 5V)

2.1 Wiring - fingerprint (CP2102)

R307S	Connect to
VCC	Pi 5V (physical pin 2) - NOT CP2102 3.3V
GND	CP2102 GND and Pi GND
TX	CP2102 RXD
RX	CP2102 TXD
CP2102 USB	Pi USB port -> /dev/ttyUSB0

If the sensor LED flashes then dies on 3.3V, that is power collapse. Always use Pi 5V for VCC.

2.2 Wiring - OLED (I2C)

OLED	Pi pin
VCC	Pin 1 (3.3V)
GND	Pin 6 (GND)
SCL	Pin 5 (GPIO3)

SDA	Pin 3 (GPIO2)
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2.3 Wiring - status LEDs

LED	Meaning	BCM GPIO	Physical
1 Net	Wi-Fi+MQTT OK / blink connecting	17	11
2 OK	Punch success (~3s)	27	13
3 Fail	Punch fail / not enrolled (~4s)	22	15

Each LED: GPIO -> 220 ohm -> LED anode -> LED cathode -> GND.

3. First-time software setup (technician)

Step 1: Flash Raspberry Pi OS, create user (e.g. taypro), connect network.

Step 2: Enable Serial (UART) and I2C in raspi-config, then reboot.

```
sudo raspi-config
# Interface Options -> Serial: login shell No, hardware Yes
# Interface Options -> I2C -> Enable
sudo reboot
```

Step 3: Clone the device repository and install Python deps.

```
cd ~
git clone https://github.com/<ORG>/device-to-erp.git
cd device-to-erp
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
cp config.example.json config.json
```

Step 4: Edit config.json (kept local, not overwritten by git pull).

```
{
  "mqtt_host": "<broker-ip>",
  "mqtt_port": 1883,
  "fingerprint_port": "/dev/ttyUSB0",
  "fingerprint_baud": 57600,
  "oled_driver": "sh1106",
  "led_net_pin": 17,
  "led_ok_pin": 27,
  "led_fail_pin": 22
}
```

Step 5: Verify sensor and OLED.

```
ls -l /dev/ttyUSB0
sudo i2cdetect -y 1          # expect 3c
python3 diagnose.py
python3 test_oled.py
```

Step 6: Install auto-start service (pull + run on every reboot).

```
python3 scripts/install_service.py
sudo reboot
```

4. HR console configuration

4.1 Device in Attendance dashboard

- Power on the Pi with network + MQTT working.
- Device publishes register/heartbeat; it appears in HR Attendance devices.
- Open the device -> set Name, Location, Latitude, Longitude.
- Save & push config (lat/lng required or punches are refused).
- Confirm device status shows Online.

4.2 Employee setup (2 fingerprints recommended)

- HR Admin -> Employees -> Add or Edit employee.
- Select the online fingerprint reader.
- Click Enroll Finger 1 -> place that finger twice when OLED says PLACE / REMOVE / PLACE AGAIN.
- Click Enroll Finger 2 -> enroll a different finger the same way.
- IDs auto-fill as FP0001 / FP0002 style values on the employee record.
- Save the employee. Either finger can punch attendance.

Do not use RFID Capture for this device. Enrollment is only via Enroll Finger buttons.

5. Daily operation (site users)

Step 1: Confirm OLED shows SCAN FINGER and top row W:OK M:OK L:OK.

Step 2: Place an enrolled finger firmly on the sensor.

Step 3: Wait for result:

- OLED: CHECK IN or CHECK OUT + employee name
- Green/OK LED briefly ON

Step 4: Lift finger before next punch. Holding the finger will not double-punch.

Step 5: If PLEASE WAIT / Finger ignored appears, wait a few seconds and try again.

5.1 OLED meanings

Screen	Meaning
SCAN FINGER	Ready for attendance
CHECK IN / CHECK OUT + name	Punch accepted
UNKNOWN FINGER	Not in sensor memory - enroll needed
NOT ENROLLED	In sensor but not linked to HR employee
PLACE / REMOVE FINGER	Enrollment coaching
PLEASE WAIT	Cooldown - too soon after last punch

5.2 LED meanings

LED	On / Blink
Net	Solid = cloud OK; Blink = connecting; Off = d
OK	Successful punch
Fail	Fail, reject, unknown finger, MQTT error

6. Software updates

Developers push code to GitHub from the laptop. On site you only reboot the Pi. On boot, boot_run.py does git reset to origin/main, installs deps, and starts main.py.

```
# After developers push:
```

```
sudo reboot
```

config.json and data/ are gitignored and survive updates. Do not edit application code permanently on the Pi.

7. Maintenance & admin tools

7.1 Flush all fingerprints on sensor

Wipes every template on the R307. HR FP IDs remain - re-enroll after.

```
source .venv/bin/activate
python3 flush_all_fingerprints.py
# type YES
```

7.2 Device logs

The device sends logs to the cloud (same MQTT a:log path as ESP). View them in HR Attendance -> device logs for this device.

7.3 Service logs on Pi

```
systemctl status taypro-fingerprint
journalctl -u taypro-fingerprint -f
tail -f ~/device-to-erp/data/boot_run.log
```

8. Troubleshooting

Symptom	Check / Fix
Sensor timeout / no header	5V VCC, TX/RX swap, /dev/ttyUSB0
OLED blank	I2C enabled, i2cdetect 0x3C, try ssd1306
W/M not OK	Network + MQTT host; device online in HR
NO LOCATION	Set lat/lng on device in HR and push
NOT ENROLLED	Enroll Finger 1/2 for that employee
UNKNOWN FINGER	Finger not on this sensor - enroll
Double punch	Update code; lift finger between punches
Logs empty in HR	Device must be registered (id+key)

9. Go-live checklist

- Hardware wired (sensor 5V, OLED I2C, 3 LEDs)
- Serial + I2C enabled; diagnose.py OK
- config.json MQTT + /dev/ttyUSB0 correct
- systemd service installed; survives reboot
- Device online in HR; lat/lng set
- Test employee enrolled with 2 fingers
- Test check-in and check-out on OLED + LEDs
- Device logs visible in HR after a punch

End of SOP. For firmware questions see device-to-erp/README.md. For HR console see taypro-console-frontend / taypro-console-backend.